

Least squares

Given $A = QR$

Then

$$A^T A \vec{x} = A^T \vec{b}$$

$$R^T \cancel{Q^T Q}^I R \vec{x} = R^T Q^T \vec{b}$$

$$R^T R \vec{x} = R^T Q^T \vec{b}$$

$$R \vec{x} = Q^T \vec{b}$$

↑
upper triangular

$$\tilde{X} = \sum_{i=1}^n \sigma_i u_i v_i^T$$

↑
pick out most important features.

Ex: Predicting cost of houses

$$\sigma, u, v$$

$$\begin{bmatrix} u_1 & u_2 & u_3 \end{bmatrix} \begin{bmatrix} \sigma_1 & & \\ & \sigma_2 & \\ & & \sigma_3 \end{bmatrix} \begin{bmatrix} v_1^T \\ v_2^T \\ v_3^T \end{bmatrix}$$

Def: basis

linearly indep set

spans $\mathbb{R}^n$

$$\vec{v} = c_1 \vec{v}_1 + \dots + c_n \vec{v}_n$$

$$\vec{v} \cdot \vec{v}_i = (c_1 \vec{v}_1 + \dots + c_n \vec{v}_n) \cdot \vec{v}_i$$

$$= c_i (\vec{v}_i \cdot \vec{v}_i)$$

$$c_i = \frac{(\vec{v}_i \cdot \vec{v})}{\|\vec{v}_i\|^2}$$

$$u_i = \frac{A \vec{v}_i}{\sigma_i}$$

$$6. \quad P^2 = P \quad P \cdot P \vec{v} = P \cdot \lambda \vec{v}$$

$$P \vec{v} = \lambda \vec{v} \quad P^2 \vec{v} = \lambda P \vec{v}$$

$$P^2 \vec{v} = \lambda P \vec{v}$$

$$P \vec{v} = \lambda P \vec{v}$$

$$\lambda = 1$$

$$8. \quad AB = BA$$

$$A \vec{v} = \lambda \vec{v}$$

$$a) \quad AB \vec{v} = BA \vec{v} \\ = B \lambda \vec{v}$$

$$\underline{AB \vec{v}} = \lambda \underline{B \vec{v}}$$

Find an orthonormal basis in $\mathbb{R}^{[0,1]}$
for the span of $\{1, 1-x^2\}$

$$\star \{f_1, f_2\} \quad [0, 1] \quad \langle f, g \rangle = \int_0^1 f(x)g(x)dx$$

$$\|1\|^2 = \int_0^1 1^2 dx = \int_0^1 1 dx = x \Big|_0^1 = 1$$

$$\|1\| = \sqrt{1} = 1$$

$$\|f\| = \sqrt{\langle f, f \rangle}$$

$$\|f\|^2 = \langle f, f \rangle$$

$$f_1^\perp(x) = 1$$

$$f_2^\perp(x) = (1-x^2) - \frac{\langle 1, 1-x^2 \rangle}{\langle 1, 1 \rangle} 1$$

$$= 1 - x^2 - \int_0^1 (1)(1-x^2) dx$$

$$= 1 - x^2 - \left[x - \frac{1}{3}x^3 \right]_0^1$$

$$= 1 - x^2 - 1 + \frac{1}{3}$$

$$= \frac{1}{3} - x^2$$

$$\|f_2^\perp\|^2 = \int_0^1 \left(\frac{1}{3} - x^2\right)^2 dx$$

$$= \int_0^1 \frac{1}{9} - \frac{2}{3}x^2 + x^4 dx$$

$$= \frac{1}{9}x - \frac{2}{9}x^3 + \frac{1}{5}x^5 \Big|_0^1$$

$$= \frac{1}{9} + \frac{1}{5} = \frac{5}{45} + \frac{9}{45} = \frac{14}{45}$$

$$\|f_2^\perp\| = \sqrt{\frac{14}{45}}$$

$$f_1^\perp(x) = 1$$

$$f_2^\perp(x) = \sqrt{\frac{45}{14}} \left(\frac{1}{3} - x^2\right)$$

$$M: \mathbb{R}^n \rightarrow \mathbb{R}^m$$

$$\mathbb{R}^n = \text{ran } M^T \oplus \ker M$$

$$\mathbb{R}^m = \text{ran } M \oplus \ker M^T$$

7.2.1

$$A = \begin{bmatrix} 3 & -2 \\ 2 & -3 \\ 1 & -1 \\ 2 & 3 \\ 3 & 4 \end{bmatrix} \quad \vec{b} = \begin{bmatrix} 2 \\ -1 \\ 0 \\ 6 \\ 0 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 3 & 2 & 1 & 2 & 3 \\ -2 & -3 & -1 & 3 & 4 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ 2 & -3 \\ 1 & -1 \\ 2 & 3 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 27 & 5 \\ 5 & 39 \end{bmatrix}$$

$$A^T \vec{b} = \begin{bmatrix} 3 & 2 & 1 & 2 & 3 \\ -2 & -3 & -1 & 3 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 0 \\ 6 \\ 0 \end{bmatrix} = \begin{bmatrix} 6 \\ -4 \end{bmatrix}$$

$$\begin{bmatrix} 27 & 5 & | & 6 \\ 5 & 39 & | & -4 \end{bmatrix}$$

$$A \vec{x} = \vec{b}$$

$$\tilde{X} \vec{\beta} = \vec{y}$$

$$\vec{x} = \begin{bmatrix} \frac{127}{514} \\ \frac{-59}{514} \end{bmatrix}$$

Find a quadratic least squares solution through the points

$$11. \quad \bar{y} = \beta_0 + \beta_1 x + \beta_2 x^2$$

$$3.2 = \beta_0 + \beta_1 \cdot 0 + \beta_2 \cdot 0^2$$

$$1.6 = \beta_0 + 0.5\beta_1 + 0.5^2\beta_2$$

$$\begin{bmatrix} 1 & x & x^2 \\ 1 & 0 & 0 \\ 1 & 0.5 & 0.25 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 2.5 & 6.25 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \\ 1 & 5 & 25 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix} = \begin{bmatrix} 3.2 \\ 1.6 \\ 2 \\ -0.4 \\ -0.8 \\ -1.6 \\ 0.3 \\ 2.2 \end{bmatrix}$$

$\tilde{X}$
 $\vec{\beta}$
 $\vec{y}$

$\sin x$
 $\cos x$

$$\begin{aligned} \vec{y} &= \beta_0 \sin x + \beta_1 \cos x \\ &= \beta_0 e^{-x} + \beta_1 \ln x \end{aligned} \left[\begin{array}{l} \sin x \\ \cos x \end{array} \right]$$

$$y = \beta_0 x^{\beta_1}$$

$$\log y = \log \beta_0 + \beta_1 x$$

$$\begin{bmatrix} 1 \\ x \end{bmatrix} \begin{bmatrix} \log \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} \log y \end{bmatrix}$$

$$\beta = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$x = \log \beta_0 \Rightarrow \beta_0 = e^x$$

$$A = PDP^{-1}$$

$$\begin{bmatrix} u_1 & u_2 & u_3 \end{bmatrix} \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \lambda_3 \end{bmatrix}$$

Each eigenvalue has one eigenvector.

$$AB = BA \quad A\vec{v} = \lambda\vec{v}$$

wish to show
 $B\vec{v} = \lambda\vec{v}$

$$AB\vec{v} = \lambda B\vec{v}$$

$$\begin{aligned} AB\vec{v} &= BA\vec{v} \\ &= \lambda B\vec{v} \end{aligned}$$

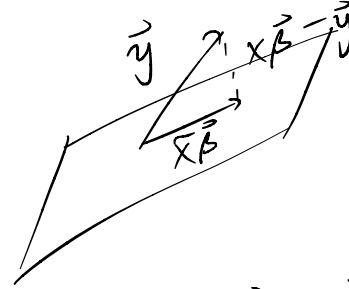
$$\tilde{X} \vec{\beta} = \vec{y}$$

↑

$$\nabla_{\vec{\beta}} \min_{\vec{\beta}} \|\tilde{X} \vec{\beta} - \vec{y}\|^2$$

$$\min_{\vec{x}} \|A \vec{x} - \vec{b}\|^2$$

$$2 \tilde{X}^T (\tilde{X} \vec{\beta} - \vec{y})$$



$$\tilde{X}^T (\tilde{X} \vec{\beta} - \vec{y}) = 0$$

$$\tilde{X}^T \tilde{X} \vec{\beta} - \tilde{X}^T \vec{y} = 0$$

$$\tilde{X}^T \tilde{X} \vec{\beta} = \tilde{X}^T \vec{y}$$